

Root Duality: Unifying Indexical Identity and Computational Anchoring

Ryan Kangjie CHANG

Institutedor Skilled Talenr Develop,ZIMET

School of Chinese Language and Literature,HNU

kjchang@hnu.edu.cn

1. Research Background and Research Questions

Within contemporary construction-based approaches to morphology, research on the *root* (ROOT) has developed along three main lines of inquiry.

(1) Scholars such as Harley (2014) regard roots as *indices*. In Harley's framework, roots are represented in the syntax as arbitrary numerical indices, serving to distinguish each root from all others. She argues that a single root may display different phonological forms in different syntactic environments. Since the semantic content of roots is highly variable—often involving a kind of “semantic substitution”—roots cannot be identified by their phonological or semantic features alone. Consequently, Harley proposes that roots should be treated in syntax as purely structural units used in syntactic computation, bearing neither semantic nor phonological features.

Building on the principled properties of *Merge*, De Belder and van Craenenbroeck (2015) further argue that roots exhibit four core characteristics:

- a.Roots bear no grammatical features;
- b.Roots have no syntactic category;
- c.Roots are defined structurally rather than lexically; and
- d.Roots are dominated by functional categories, rather than the other way around.

(2) In contrast, Marantz (1997), Harley (2009a), and Embick (2004a) assume that roots possess inherent ontological properties. Marantz distinguishes *entity roots* (e.g., objects), *state roots* (e.g., properties), and *event roots* (e.g., actions). Similarly, Rappaport Hovav & Levin (2010) divide roots into *manner* and *result* types, proposing that a verb's syntactic behavior is determined by the semantic type of its root.

(3) A third line of thought, represented by Acedo-Matellán & Mateu (2015) and Borer (2013), maintains that the semantic interpretation of roots is entirely derived from syntactic structure rather than from any inherent lexical meaning. Roots are *category-free* abstract units whose interpretation depends on the functional heads (such as *v*, *n*) with which they combine.

As mentioned above, Harley's theory treats roots as meaningless, phonologically null indices. However, Marantz (2024)¹ points out that indices such as “#53” or “#67” are not elements actually manipulated by syntactic computation; rather, the *root* serves as a *computational anchoring*. Morphological alternations like *go* → *went* begin from this base. Therefore, we can posit that the underlying form of the root should be phonetic in nature, which serves as the foundational ontological framework introduced in the preceding discussion.

Regarding the “meaningless” approach, Borer (2013), in her *Exo-Skeletal Model*, contends that roots have no inherent meaning; instead, their conceptual content is retrieved at a later stage of the derivation, when the syntax accesses encyclopedic knowledge.

To summarize, current linguistic theories differ in how they conceptualize the nature of roots, reflecting divergent empirical and theoretical priorities. Regarding the ontological status of roots, I follow De Belder & van Craenenbroeck (2015) in proposing that the *first Merge operation* in each cyclic domain creates a radically empty structural position at the base of the structure, into which a root can be inserted at the stage of Vocabulary Insertion. I further argue that the root's function in linguistic generation is *dual and phase-dependent*: during the derivation from syntax to the two interfaces—Sensory-Motor (SM) and Conceptual-Intentional (CI)—the root functions both as an *indexical unit* and as a *computational anchoring*.

Building upon De Belder & van Craenenbroeck's insight, this paper reexamines the properties of roots and proposes the Dual Nature of the Root. It further explores how this duality constrains the derivation of compound word

¹ Academic Correspondence between the Author and Alec Marantz.

formation. The remainder of this paper is structured as follows: Section 2 presents the hypothesis of root duality; Section 3 discusses how the root's indexical and computational properties shape the derivation of compounds; Section 4 outlines the theoretical implications of root duality for compound formation; and Section 5 concludes the paper.

2. What Is a Root?

The question “What is a root?” constitutes the conceptual premise for any discussion of the root's properties.

According to De Belder and van Craenenbroeck (2015), within the frameworks of the Minimalist Program (MP) and Distributed Morphology (DM), roots are structurally defined rather than lexically defined. They propose the notion of Asymmetric Primary Merge, arguing that syntactic building is inherently directional: the Primary Merge always combines a set of grammatical features (e.g., $\{\alpha\}$) with the empty set ($\emptyset$), forming an ordered pair $\langle\{\alpha\}, \emptyset\rangle$, with $\{\alpha\}$ serving as the projecting and labeling element. The Primary Merge thus creates an empty structural position at the base of the derivation—precisely the position into which a root is later inserted at the stage of Vocabulary Insertion.

On this basis, De Belder and van Craenenbroeck (2015) regard the root as a syntactic “empty structural position” devoid of inherent grammatical information or categorial features, which only receives a syntactic category through the attachment of a functional head. They further demonstrate that roots possess the following four properties:

- (1) roots have no grammatical features;
- (2) roots bear no syntactic category;
- (3) roots are defined by syntactic structure, not by the lexicon; and
- (4) roots are dominated by functional heads, rather than the reverse.

We aligns with De Belder and van Craenenbroeck's analysis and further argues that the ROOT functions as both an index at the interface and as a phase-

dependent computational unit. During the syntactic derivation stage—where List 1 provides the syntactic terminals for computation—the root occupies an empty position, as described by De Belder and van Craenenbroeck (2015): “The first Merge operation in each cyclic domain—called Primary Merge here—creates a radically empty structural position at the foot of the structure in which a root can be inserted at the level of Vocabulary Insertion.”

We further propose that, following syntactic Merge and agreement operations, this empty position serves as an interface marker for subsequent insertion, functioning as an indexical unit. During the morphological operation stage—when lexical items are provided from List 1—this indexical unit undergoes ordered linearization. At this stage, the root (ROOT) functions as the baseline for morphological computation, meaning that its phonologically realized base form provides the reference point for both morphological and phonological interface operations; morphological modifications are carried out accordingly (List 2 provides the vocabulary for this stage). Similarly, at the encyclopedic-knowledge stage—when information is drawn from List 3—the root also serves as the computational anchoring for semantic operations, as illustrated in Figure 1:

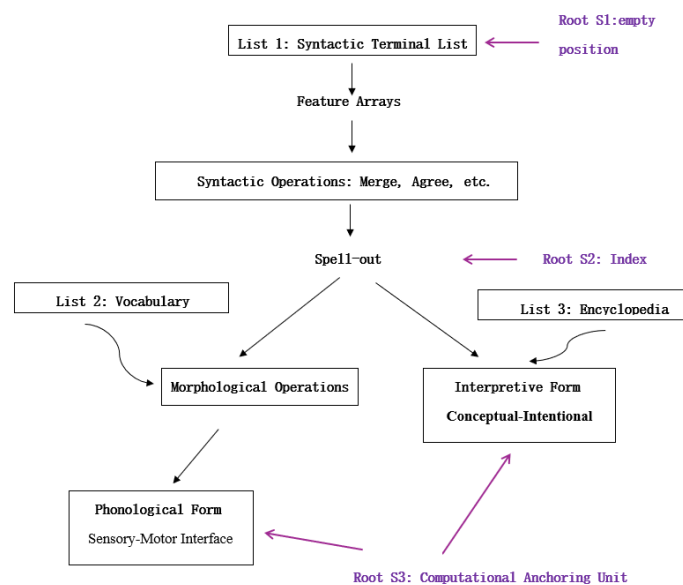


Figure 1 schematically represents this phase-based derivational process.

In summary, the present study argues that the root (ROOT) is a phase-dependent unit exhibiting a dual nature: it simultaneously functions as an indexical identifier and as a computational anchoring, unifying the interpretive and phonological aspects of the derivation within a single syntactic architecture.

3. The Indexical Function and Computational Anchoring of the Root

During the process of syntactic building, the stage of Primary Merge gives rise to what can be termed a Null Derivation—that is, at the initial point of syntactic construction, no substantive lexical items have yet been introduced. The derivation begins with a Merge operation involving the empty set ($\emptyset$), resulting in a structure that is, at this early stage, *empty* of both phonological and semantic content.

This *empty position* constitutes what may be understood as an index: once word formation is completed, this position is transferred to the interface, where it receives phonological and semantic interpretation. Following Marantz (2007), words themselves can be treated as phases, and word formation accordingly obeys the Phase Impenetrability Condition (PIC), ensuring that each morphological domain is independently interpreted and transferred.

Within the syntactic building process, the *empty position*—which we may refer to as the initial root (ROOT_c)—exhibits two structural properties derived from Merge: Operability ([OY]) and Baseline Startup Capability ([BSC]).

a. Operability enables the initial root to participate in combinatorial operations, such as binary Merge with another root or with a categorizing head, thereby initiating word formation.

b. Baseline Startup Capability, in turn, characterizes the post-syntactic status of this empty position: once syntactic computation is completed, the position requires interface-level insertion of phonological and semantic content. At this point, the root becomes the computational anchoring for both morphophonological and semantic interpretation, functioning as an index (indices) recognizable at the interfaces.

To illustrate this, consider two examples of compound formation in Chinese:

“da huang gou” (大黄狗, big yellow dog) and “huang da yi” (黄大衣, yellow coat). These two types of compounding display distinct structural configurations, yet both instantiate the same fundamental mechanism in which an initially empty root position—endowed with operability and baseline startup capability—serves as the site of subsequent insertion and interface identification.

(1) $[\# \sqrt{\text{huang}} [\text{nP} < \sqrt{\text{da}} \sqrt{\text{yi}} >, \text{n}]]$

$[\# \sqrt{\text{黄}} [\text{nP} < \sqrt{\text{大}} \sqrt{\text{衣}} >, \text{n}]]$

yellow coat

(2) $<[\# \sqrt{\text{da}} [\sqrt{\text{haung}} \sqrt{\text{gou}}]], \text{n}>$

$<[\# \sqrt{\text{大}} [\sqrt{\text{黄}} \sqrt{\text{狗}}]], \text{n}>$

big yellow dog

In the morphological analysis, (1) *huang* + *dayi* (“yellow coat”) represents an outer-domain derivation, in which [*dayi*] functions as a phasal constituent that is directly transferred at the interface. In contrast, (2) *da huang gou* (“big yellow dog”) constitutes an inner-domain derivation, where *da* and *huang* are adjoined within the root domain. Due to the Phase Impenetrability Condition (PIC), the constituent *dayi* in (1) can no longer undergo further syntactic operations once it has been transferred.

At the first stage of root derivation (Stage S1 in Figure 1), the roots *huang*, *da*, *yi* and *da*, *huang*, *gou* are listed in List 1. We assume that during Primary Merge, the initial root (ROOTc) is merged with a categorizing morpheme (n). At this point, guided by the two structural properties—Operability ([OY]) and Baseline Startup Capability ([BSC])—the following structural changes take place:

(1a)

Primary Merge: $[\{\emptyset, \emptyset\}, \text{n}] \longrightarrow <\{\emptyset, \emptyset\}, \text{n}>$ **【OY】**

Merge-and-sequential-labeling stage: $\{\{\emptyset, \emptyset\}, \text{n}\} \longrightarrow <\{\emptyset_1, \emptyset_2\}, \text{n}>$

Structure Building: $[\{\emptyset\}, <\{\emptyset_1, \emptyset_2\}, \text{n}>] \longrightarrow \{\{\emptyset\}, <\{\emptyset_1, \emptyset_2\}, \text{n}>\}$

Derivation by Phase: $\{\{\emptyset\}, <\{\emptyset_1, \emptyset_2\}, \text{n}>\} \longrightarrow \{\{\emptyset\}, <\{\emptyset_1, \emptyset_2, \text{n}>\}$ **【BSC】**

At this point, the phasal unit $<\{\emptyset_1, \emptyset_2\}, \text{n}>$ is transferred to the interface for further operations (corresponding to Stages S2 and S3 in Figure 1).

Merge-and-sequential-labeling stage: $\{\{\emptyset\}, \langle \{\emptyset_1, \emptyset_2\}, n \rangle\} \longrightarrow \{\{\emptyset_3\}, nP\}$

Derivation by Phase: $\{\{\emptyset_3\}, nP\}$ **【BSC】**

At this stage, the word-forming phase is derived up to its highest node, and the phasal unit $\{\{\emptyset_3\}, nP\}$ is subsequently transferred to the interface, completing the process of word formation.

In cases of coordinate-type compounds such as *dayi* (“coat”), where the Primary Merge introduces two empty positions that merge with a lexically realized morpheme, I argue that this arises from the Operability feature ([+OP]) inherent to the initial root. This feature activates the empty root position, making it available for further merger with other elements. Since [+OP] is a syntactic structural feature rather than a categorical one, it cannot bear a label by itself. Consequently, when the root merges with a lexicalized morpheme, the categorical feature of the latter is shared and serves as the label for the merged structure. However, when the root merges with another initial root that also carries [+OP], the resulting empty position—referred to here as an Operability Null Set (OperNS)—contains two component positions (as in the case of *dayi*), as illustrated in (1b)².

$$(1b) \quad \{\emptyset_1\}, \{\emptyset_2\} \longrightarrow \{\emptyset_1, \emptyset_2\} = \text{OperNS} \\ [\text{OperNS}, n] \longrightarrow \langle \{\emptyset_1, \emptyset_2\}, n \rangle$$

It should be noted that the subscripts (1, 2, 3, etc.) assigned to the empty positions merely indicate the order of merger, rather than serving as identificational indices. Therefore, this stage is referred to as the Merge-and-Sequential-Labeling Stage.

At the $\{\{\emptyset_3\}, nP\}$ stage, the merger of $\emptyset_3$ with nP triggers the Transfer of the syntactic structure to the interface. During the mapping to the sensorimotor (SM) interface, the index of the initial root and the features of the categorizing morpheme jointly activate the Baseline Startup Capability (BSC). This activation initiates the search, within List 2 (Vocabulary), for the corresponding Vocabulary Items (VIs) to

² What we need to clarify is that the operable null set (OperNS) here is actually the empty position in the syntactic structure. The reason we set up this set is determined by the function of the indexing unit of the initial root.

be inserted. Simultaneously, the conceptual-intentional (C-I) system accesses encyclopedic knowledge to assign semantic interpretation.

The formation of compounds such as *dahuanggou* (“big yellow dog”) follows the same principle: an Operability Null Set (OperNS) merges with a categorizing morpheme. The distinction lies in the fact that, in this case, three initial roots are involved. The first two merge to form OperNS₁, which subsequently merges with the third root to yield OperNS₂. Finally, this complex OperNS₂ merges with the categorizing morpheme *n*, as illustrated in (2a).

$$\begin{aligned}
 (2b) \quad & \{\emptyset_1\}, \{\emptyset_2\} \longrightarrow \{\emptyset_1, \emptyset_2\} = \text{OperNS}_1 \\
 & \{\emptyset_3\}, \text{OperNS}_1 \longrightarrow \{\emptyset_3, \text{OpeP}_1\} = \{\emptyset_3, \{\emptyset_1, \emptyset_2\}\} = \text{OperNS}_2 \\
 & [\text{OperNS}_2, n] \longrightarrow \langle \{\emptyset_3, \{\emptyset_1, \emptyset_2\}\}, n \rangle
 \end{aligned}$$